rusEFI STI 2004: plug-and-play ECU for the 2004 USDM Subaru Impreza WRX STI (EJ257, drive-by-wire).

Built from Hellen-One modules. Signals between sheets are global net labels; power symbols are global.
Crossed-out (DNP) parts are optional footprints, not assembled: what each one is for is in documents/README.md.
Car pinout and cross-check: documents/STI2004-pinout-check.xlsx (tools/pinout_check.py).
Sheet order follows the signal: car connector -> power -> MCU -> inputs -> outputs -> comms -> aux.

Vehicle connector

File: Connector.kicad_sch

Units: D = B137, C = B136, B = B135, A = B134. Every pin carries a global net label used on the other sheets.

Power

File: Power.kicad_sch

M14 (main power module): +12V_RAW from the main relay (B5/B6), +12V_IGN (D15), back-up +12V (B19).

MCU and USB

File: Mcu.kicad_sch

M1: Hellen mega-mcu144 (STM32F7), with a 120R CAN terminator built in.

Sensors

File: Sensors.kicad_sch

Crank: 2-wire VR sensor (B10 +, B22 -) into M5. Knock (C25) into M13. M7 is a spare knock channel (not used).

Switch inputs

File: Inputs.kicad_sch

M11/M12 (dinput4): car lines that close to ground (defogger D11, lights D12, blower D13, test mode D14,

Outputs

File: Outputs.kicad_sch

Low-side drivers (NCV8401B, protected) in sub-sheets: INJ injectors 1-4; LS1 main relay, fans, ETB relay;

Comms

File: Comms.kicad_sch

M3: rusEFI wideband (WBO) for a Bosch LSU 4.9 front O2, on CAN to M1. It reaches the sensor through the stock

Aux and expansion

File: Aux.kicad_sch

J3: Molex Mini-Fit Jr 2x7 (pinout in documents/README.md): fused 12 V and 5 V, sensor ground, three digital

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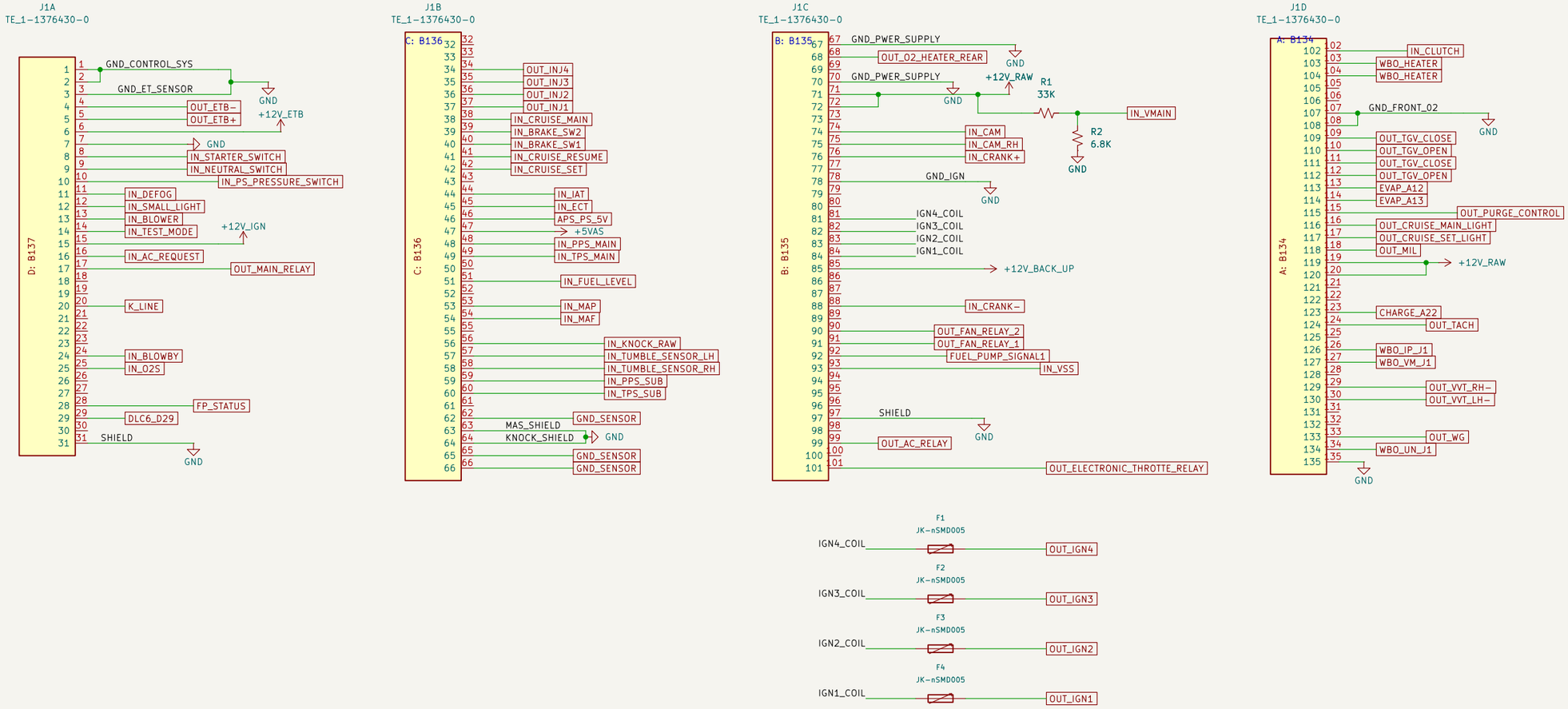
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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: rusefiSTI2004	PART NUMBER: TBD		REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 1 OF 14

Vehicle connector J1 (TE 1-1376430-0): the stock 2004 STI ECM plugs.

Units: D = B137, C = B136, B = B135, A = B134. Every pin carries a global net label used on the other sheets.
Pin functions: documents/oem-ecm-pinout-2004-sti.csv (FSM EN(STi)-18..22, wiring WI-106..118); check: tools/pinout_check.py.
Coil outputs B15-B18 run through F1-F4 (50 mA PTC, block below J1C, joined by IGNn_COIL labels). R1/R2 divide battery voltage for IN_VMAIN.



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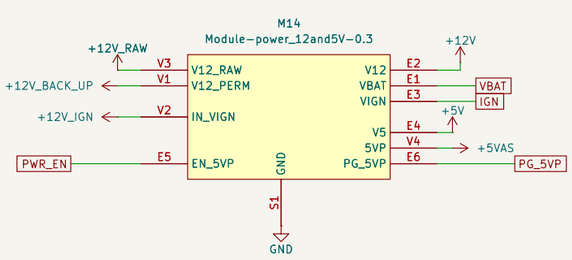
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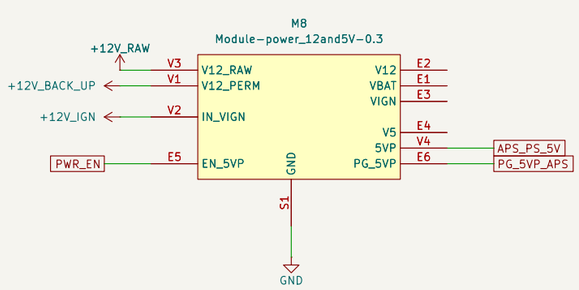
EMBEDDEDERA

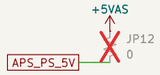
TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: Vehicle connector	PART NUMBER: TBD		REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 2 OF 14

Power.

M14 (main power module): +12V_RAW from the main relay (B5/B6), +12V_IGN (D15), back-up +12V (B19).
It makes +12V, +5V and +5VAS, the protected sensor 5 V for throttle TPS, MAP and TGV sensors (C16).
M8 (second power module) feeds only the pedal 5 V (APS_PS_5V, C15), so pedal and throttle sensors are on independent rails and their plausibility check works. The open link puts the pedal on +5VAS if M8 is not fitted.









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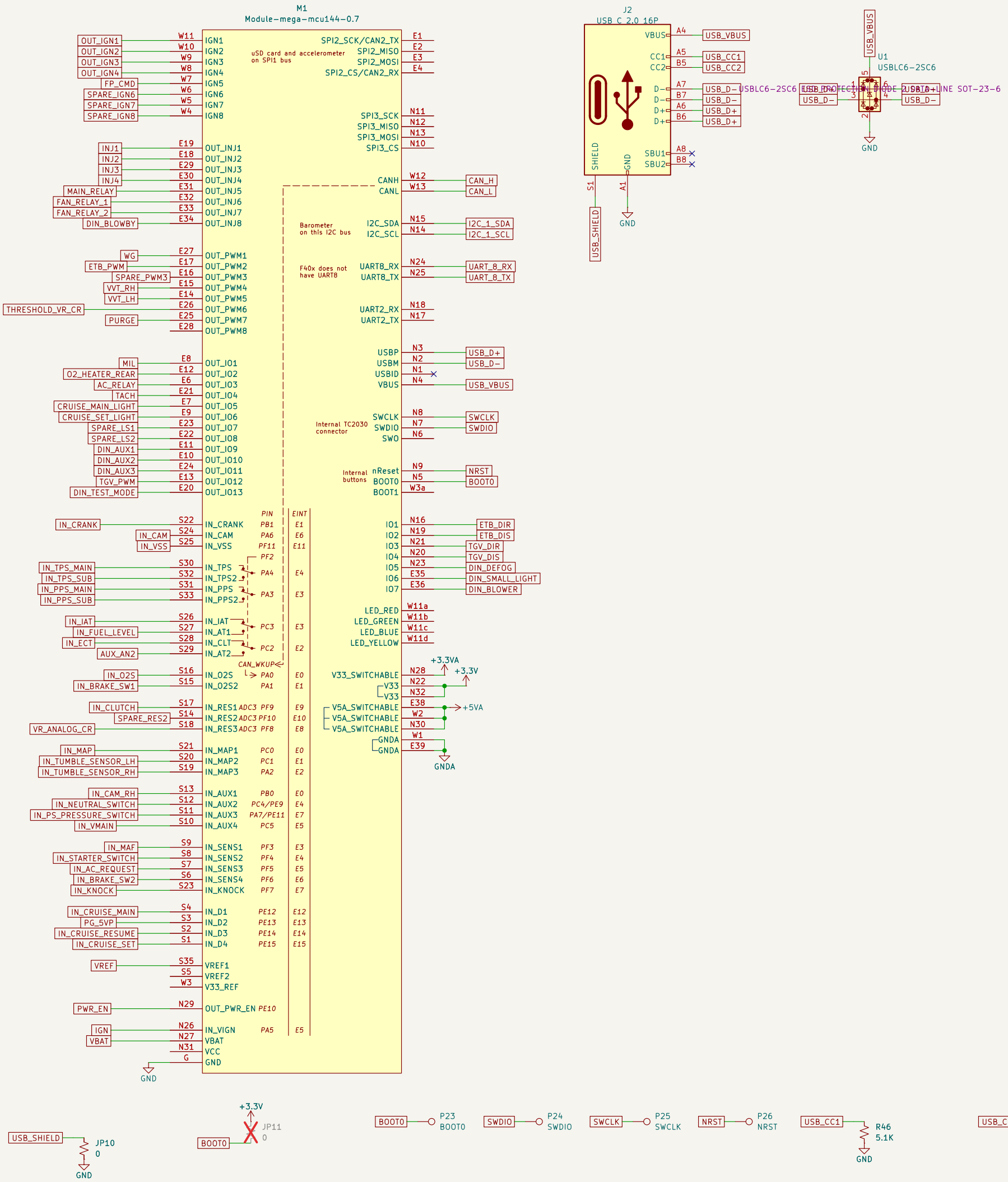
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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: Power		PART NUMBER: TBD	REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 3 OF 14

MCU and USB.

M1: Hellen mega-mcu144 (STM32F7), with a 120R CAN terminator built in.
J2 USB-C for TunerStudio: 5.1k CC pull-downs (device), USBLC6-2SC6 ESD (U1), shell to GND through a 0R link.
BOOT0 pad with an open link to 3.3 V forces the USB DFU bootloader; SWD pads (SWDIO, SWCLK, NRST) for a debugger.



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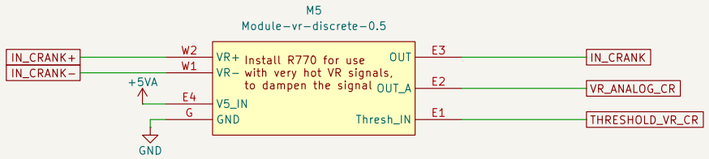
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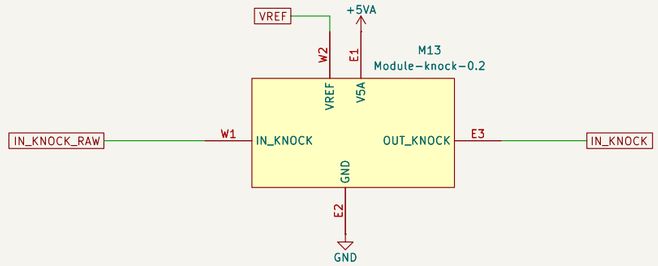
EMBEDDEDERA

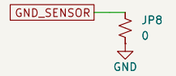
TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: MCU and USB	PART NUMBER: TBD		REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 4 OF 14

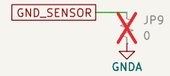
Engine sensors.

Crank: 2-wire VR sensor (B10 +, B22 -) into M5. Knock (C25) into M13 (the STI has one knock sensor).
Cams: 3-wire Hall sensors (WI-114), wired straight to M1 inputs that have pull-ups; crossed parts = stronger pull-ups.
Analog sensors (MAF, MAP, IAT, ECT, TPS, pedal, TGV, fuel level) go straight to M1 inputs by net label.
Sensor returns C31/C34/C35 (GND_SENSOR): to GND through the fitted 0R, or to GNDA through the open link.













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EMBEDDEDERA

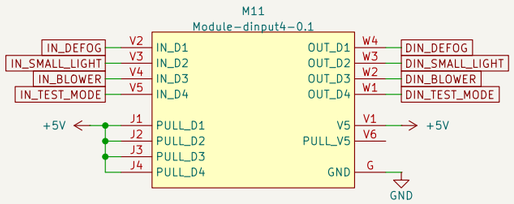
TITLE: rusEFI STI 2004 ECU | COSTUMER: EmbeddedEra

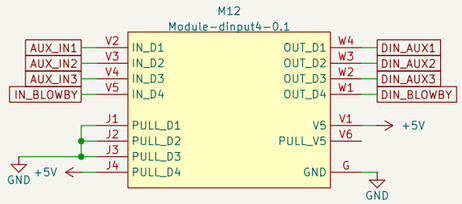
DESCRIPTION: Sensors | PART NUMBER: TBD | REV: B

SIZE: A3 | DATE ISSUE: 2026-10-13 | DESIGNER: AGD | SHEET: 5 OF 14

Switch inputs.

M11/M12 (dinput4): car lines that close to ground (defogger D11, lights D12, blower D13, test mode D14, blow-by D24) with 680k pull-ups, and three aux inputs (launch / 2-step, map switch, spare) that expect 12 V. 12 V-active switches (starter, A/C request, brake 1 and 2, clutch) and the cruise buttons are read on M1 analog inputs with the fitted pull-downs here. Crossed-out parts are options: see documents/README.md.

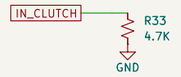


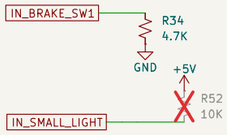


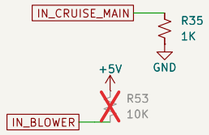






















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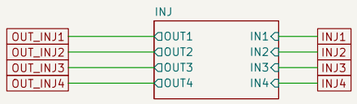
TITLE: rusEFI STI 2004 ECU | COSTUMER: EmbeddedEra

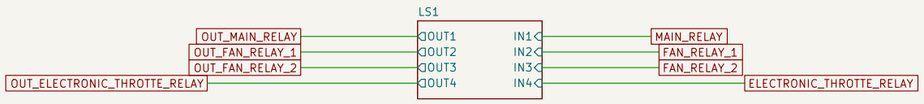
DESCRIPTION: Switch inputs | PART NUMBER: TBD | REV: B

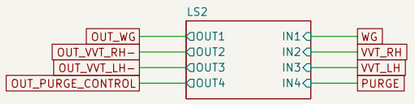
SIZE: A3 | DATE ISSUE: 2026-10-13 | DESIGNER: AGD | SHEET: 6 OF 14

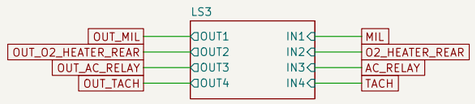
Outputs.

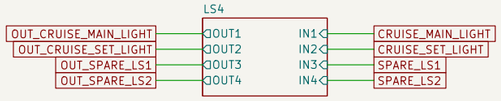
Low-side drivers (NCV8401B, protected) in sub-sheets: INJ injectors 1-4; LS1 main relay, fans, ETB relay;
LS2 wastegate, AVCS RH/LH, purge; LS3 check-engine lamp, rear O2 heater, A/C relay, tach;
LS4 cruise lamps and two spares (aux connector, or EVAP A12/A13 through the open links).
M2 drives the electronic throttle motor; M9 drives both TGV motors in parallel.
Fuel pump control unit command B26: 5 V PWM from IGN5 through 100R, with pull-up/down options.

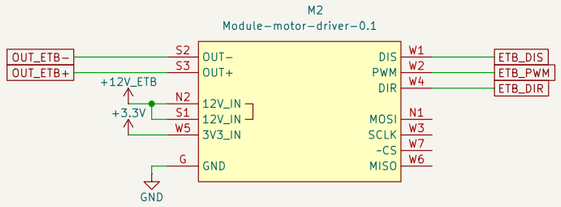


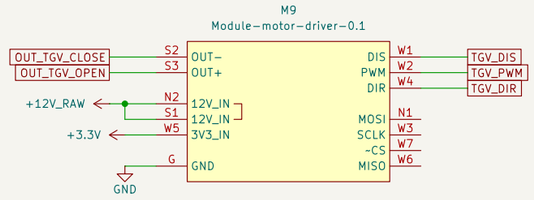








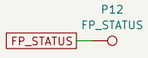


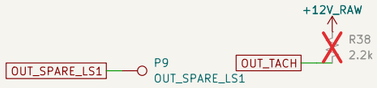




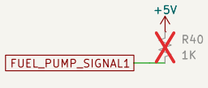












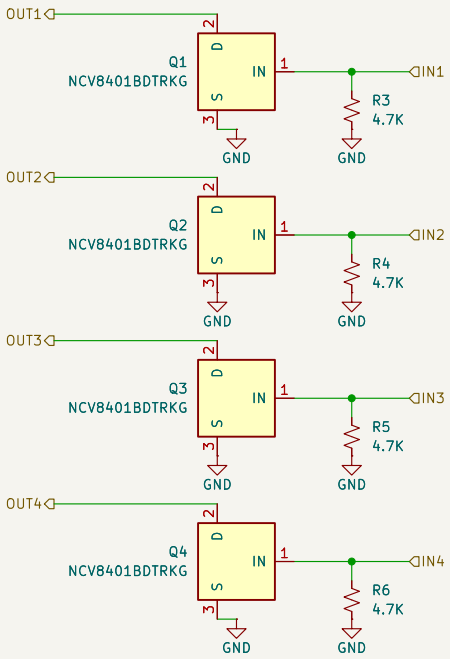


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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: Outputs		PART NUMBER: TBD	REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 7 OF 14

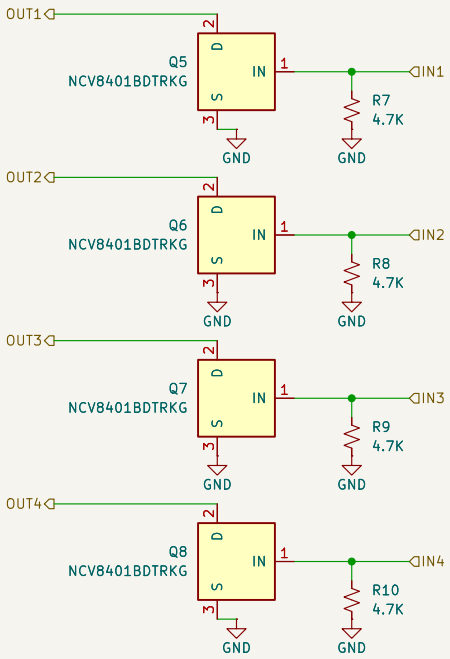


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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU		COSTUMER: EmbeddedEra	
DESCRIPTION: INJ	PART NUMBER: TBD		**REV: B**
SIZE: A4	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 10 OF 14

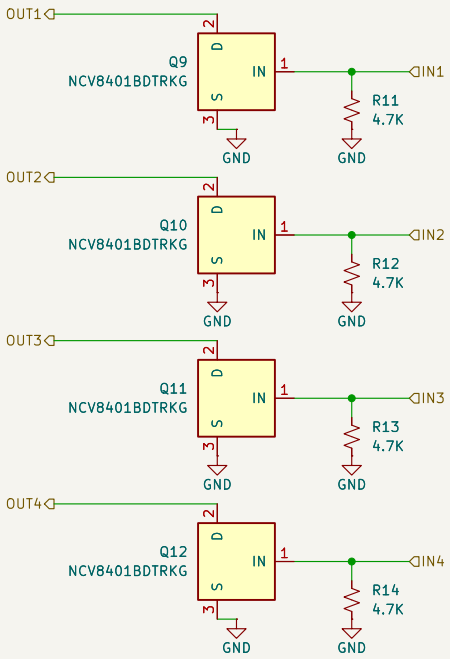


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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU		COSTUMER: EmbeddedEra	
DESCRIPTION: LS1	PART NUMBER: TBD		**REV: B**
SIZE: A4	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 11 OF 14

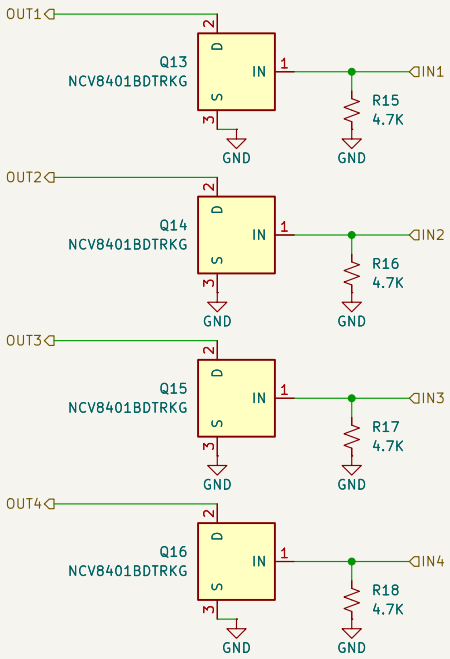


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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU		COSTUMER: EmbeddedEra	
DESCRIPTION: LS2	PART NUMBER: TBD		**REV: B**
SIZE: A4	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 12 OF 14

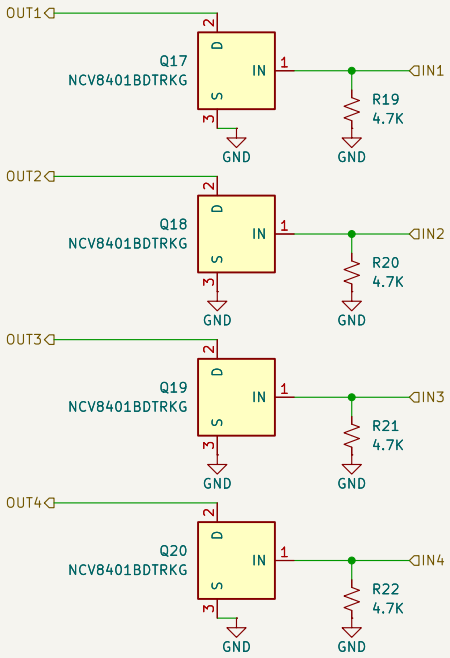


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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU		COSTUMER: EmbeddedEra	
DESCRIPTION: LS3	PART NUMBER: TBD		**REV: B**
SIZE: A4	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 13 OF 14



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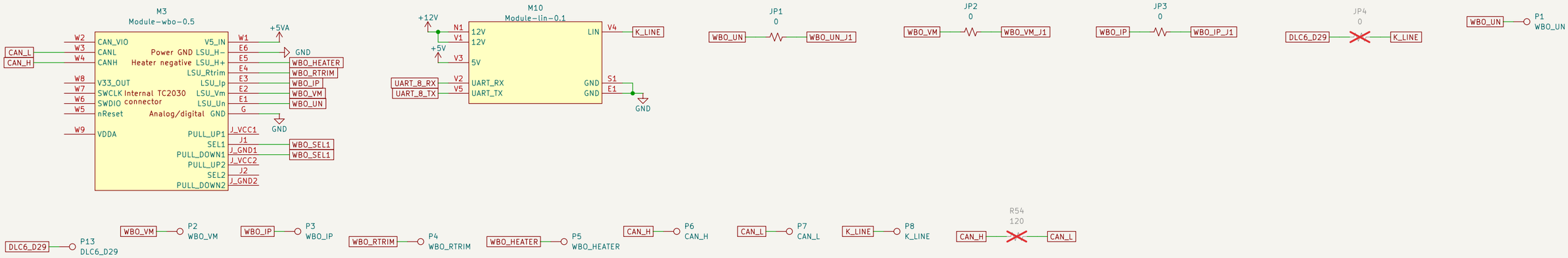
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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: LS4	PART NUMBER: TBD		**REV: B**
SIZE: A4	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 14 OF 14

Communications.

M3: rusEFI wideband (WBO) for a Bosch LSU 4.9 front O2, on CAN to M1. It reaches the sensor through the stock harness (A33/A26/A25, heater A2+A3) via the 0R links, or through a pigtail on the pads (adapter table in README).
M10: LIN transceiver used as the K-line (SSM, D20). D29 reaches the K-line only through an open link.
CAN H/L on pads, with an optional 120R terminator for an external bus end.



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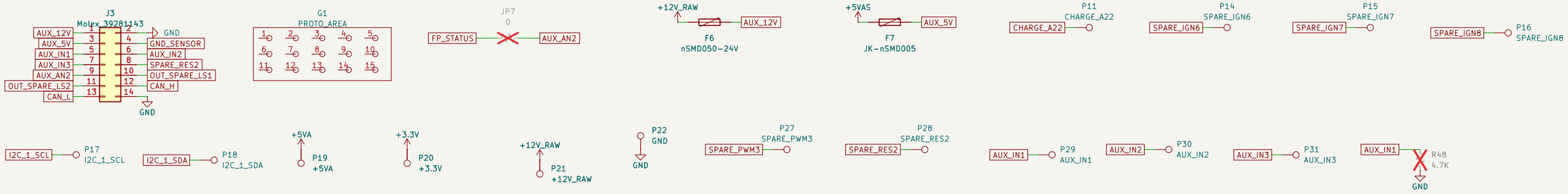
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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: Comms		PART NUMBER: TBD	REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 8 OF 14

Aux connector and expansion.

J3: Molex Mini-Fit Jr 2x7 (pinout in documents/README.md): fused 12 V and 5 V, sensor ground, three digital inputs, two analog inputs, two low-side outputs, CAN.
Spare MCU pins (IGN6-8, I2C, OUT_PWM3, IN_RES2) and every rail are on pads next to the prototype area G1.



Mounting: H1-H4 M3 plated holes, no net (the board is not grounded to the case through them).
J1's three screws hold the connector to the board only.

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EMBEDDEDERA

TITLE: rusEFI STI 2004 ECU			COSTUMER: EmbeddedEra
DESCRIPTION: Aux and expansion		PART NUMBER: TBD	REV: B
SIZE: A3	DATE ISSUE: 2026-10-13	DESIGNER: AGD	SHEET: 9 OF 14